

Math 162

Problem set 2 solutions

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1. (a) As suggested in the hint, we first note that if P is an equipartition of $[a, b]$ into subintervals of width $\Delta x = \frac{1}{n}(b - a)$, then

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{i=1}^n (M_i - m_i) \Delta x \\ &= \sum_{i=1}^n (f(x_i) - f(x_{i-1})) \Delta x \\ &= (f(x_n) - f(x_0)) \Delta x \\ &= (f(b) - f(a)) \Delta x. \end{aligned}$$

Here we used the observation that $M_i = f(x_i)$ and $m_i = f(x_{i-1})$ for any $1 \leq i \leq n$ because f is increasing (the maximum and minimum on each subinterval occur at the right and left endpoints, respectively).

Given $\epsilon > 0$, choose $n \in \mathbb{N}$ so that $n > \frac{1}{\epsilon}(b - a)(f(b) - f(a))$. Then we have

$$U(f, P) - L(f, P) = (f(b) - f(a)) \Delta x = (f(b) - f(a)) \cdot \frac{1}{n}(b - a) < \epsilon,$$

so the Darboux criterion implies that f is integrable.

- (b) Consider the following piecewise constant function $f : [0, 1] \rightarrow \mathbb{R}$. For any $n \in \mathbb{N}$, we define $f(x) := \frac{n-1}{n}$ if $\frac{n-1}{n} \leq x < \frac{n}{n+1}$, and we put $f(1) := 1$. (Try writing out the first few cases $n = 1, 2, 3$ to get a feel for what f looks like.) If $m, n \in \mathbb{N}$ satisfy $m < n$, then

$$\frac{m-1}{m} = 1 - \frac{1}{m} < 1 - \frac{1}{n} = \frac{n-1}{n},$$

and certainly $\frac{n-1}{n} = 1 - \frac{1}{n} < 1$ for any $n \in \mathbb{N}$. This implies that f is increasing (in between the cutoff points $\frac{n}{n+1}$, it is constant). On the other hand, for any $n \in \mathbb{N}$ we have

$$\lim_{x \rightarrow \frac{n}{n+1}^-} f(x) = \frac{n-1}{n} \neq \frac{n}{n+1} = \lim_{x \rightarrow \frac{n}{n+1}^+} f(x),$$

so f is not continuous at $x = \frac{n}{n+1}$. This shows that f has infinitely many discontinuities.

2. Let P_n be the partition of $[0, b]$ into n subintervals of equal length. Note that x^3 is an increasing function (this is already enough to see that it is integrable by 1(a), but we will also show it directly in the course of this calculation). In particular we have

$$M_i = x_i^3 = \left(\frac{ib}{n}\right)^3 \quad \text{and} \quad m_i = x_{i-1}^3 = \left(\frac{(i-1)b}{n}\right)^3$$

for any $1 \leq i \leq n$. Now compute

$$U(f, P_n) = \sum_{i=1}^n M_i \Delta x = \sum_{i=1}^n \left(\frac{ib}{n}\right)^3 \cdot \frac{b}{n} = \frac{b^4}{n^4} \sum_{i=1}^n i^3,$$

and similarly

$$L(f, P_n) = \sum_{i=1}^n m_i \Delta x = \sum_{i=1}^n \left(\frac{(i-1)b}{n} \right)^3 \cdot \frac{b}{n} = \frac{b^4}{n^4} \sum_{i=1}^n (i-1)^3 = \frac{b^4}{n^4} \sum_{i=1}^{n-1} i^3.$$

In particular, we see that

$$U(f, P_n) - L(f, P_n) = \frac{b^4}{n^4} \cdot n^3 = \frac{b^4}{n}.$$

Given $\epsilon > 0$, if we choose $n \in \mathbb{N}$ so that $n > \frac{b^4}{\epsilon}$, then

$$U(f, P_n) - L(f, P_n) = \frac{b^4}{n} < \epsilon.$$

Thus f is integrable on $[0, b]$ by the Darboux criterion. To see that $\int_0^b x^3 dx = \frac{1}{4}b^4$, we first observe that

$$U(f, P_n) = \frac{b^4}{n^4} \sum_{i=1}^n i^3 = \frac{b^4}{n^4} \cdot \frac{1}{4} n^2 (n+1)^2 = \frac{b^4(n+1)^2}{4n^2}$$

by the suggested identity, and similarly

$$L(f, P_n) = \frac{b^4}{n^4} \sum_{i=1}^{n-1} i^3 = \frac{b^4}{n^4} \cdot \frac{1}{4} (n-1)^2 n^2 = \frac{b^4(n-1)^2}{4n^2}.$$

Thus

$$L(f, P_n) = \frac{b^4(n-1)^2}{4n^2} < \frac{b^4}{4} < \frac{b^4(n+1)^2}{4n^2} = U(f, P_n)$$

for any $n \in \mathbb{N}$. Since

$$L(f, P_n) \leq \int_0^b x^3 dx \leq U(f, P_n)$$

as well for any $n \in \mathbb{N}$, it follows that if we fix $\epsilon > 0$ and choose $n > \frac{b^4}{\epsilon}$ as above, then

$$\left| \int_0^b x^3 dx - \frac{b^4}{4} \right| \leq U(f, P_n) - L(f, P_n) < \epsilon.$$

Since ϵ was arbitrary, the desired identity follows.

3. (a) As suggested, we first consider the case where g is the constant function 0. A constant lower bound for f gives rise to a lower bound for its integral (Spivak Chapter 13, Theorem 7): if $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f(x) dx \geq 0 \cdot (b-a) = 0.$$

In general, if $f(x) \geq g(x)$ for all $x \in [a, b]$, then $f(x) - g(x) \geq 0$ and hence

$$\int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b (f(x) - g(x)) dx \geq 0.$$

This is equivalent to the desired inequality.

- (b) First, we claim that there exist $m > 0$ and $\delta > 0$ such that for any $x \in [a, b]$ satisfying $|x - c| \leq \delta$, we have $f(x) \geq m$. For example, we can take $m := \frac{1}{2}f(c)$. Applying the definition of continuity at c with $\epsilon = m$, there exists $\delta' > 0$ such that for any $x \in [a, b]$ satisfying $|x - c| < \delta'$, we have $|f(x) - f(c)| < m$. Note that $|f(x) - f(c)| < m$ implies that

$$f(x) > f(c) - m = \frac{1}{2}f(c) = m.$$

Taking $\delta := \frac{1}{2}\delta'$, the claim follows. We can finish the proof in two different ways.

Using part (a): suppose first that $a < c < b$, so without loss of generality we can assume that $a < c - \delta$ and $c + \delta < b$ by shrinking δ if necessary. Then we have

$$\int_a^b f(x) dx = \int_a^{c-\delta} f(x) dx + \int_{c-\delta}^{c+\delta} f(x) dx + \int_{c+\delta}^b f(x) dx \geq \int_{c-\delta}^{c+\delta} f(x) dx \geq m \cdot 2\delta > 0.$$

Here the first inequality used the bound $f(x) \geq 0$ for $x \in [a, b]$, and the second inequality used the bound $f(x) \geq m$ for $x \in [c - \delta, c + \delta]$. If $c = a$, then shrink δ if necessary so that $a + \delta < b$, and then we have

$$\int_a^b f(x) dx = \int_a^{a+\delta} f(x) dx + \int_{a+\delta}^b f(x) dx \geq \int_a^{a+\delta} f(x) dx \geq m \cdot \delta > 0.$$

The case $c = b$ is similar. Note that in the end we did not use the full strength of part (a), only the basic bound on integrals.

From the definitions: it suffices to show that for some partition P of $[a, b]$, we have $L(f, P) > 0$, since then it will follow that

$$\int_a^b f(x) dx \geq L(f, P) > 0.$$

For this, choose P so that for some $1 \leq j \leq n$, we have $[x_{j-1}, x_j] \subset [c - \delta, c + \delta]$. Certainly $m_i \geq 0$ for all $1 \leq i \leq n$ because $f(x) \geq 0$ for all $x \in [a, b]$, while $f(x) \geq m$ for $x \in [c - \delta, c + \delta]$ implies that $m_j \geq m > 0$. Thus

$$L(f, P) = \sum_{i=1}^n m_i \Delta x_i \geq m_j \Delta x_j > 0$$

as claimed.

4. (a) Since f is integrable, we know that $f + g$ and $f - g$ are integrable (sums and scalar multiples of integrable functions are integrable). If we can prove that $(f + g)^2$ and $(f - g)^2$ are integrable, then it will similarly follow that

$$fg = \frac{1}{4}((f + g)^2 - (f - g)^2)$$

is integrable. Thus it suffices to prove that if f is integrable, then f^2 is integrable.

For this, we follow the hint and observe that for any $x, y \in [a, b]$, we have

$$\begin{aligned} |f(x)^2 - f(y)^2| &= |f(x) + f(y)| \cdot |f(x) - f(y)| \\ &\leq (|f(x)| + |f(y)|) |f(x) - f(y)| \\ &\leq 2M |f(x) - f(y)|. \end{aligned}$$

Give a partition P of $[a, b]$, we use the notation M'_i, m'_i for the supremum and infimum, respectively, of f^2 on $[x_{i-1}, x_i]$ (reserving M_i, m_i for the supremum and infimum of f on the same interval). We claim that the above inequality implies that

$$M'_i - m'_i \leq 2M(M_i - m_i).$$

Namely, if this were false then there would exist $x, y \in [x_{i-1}, x_i]$ such that

$$|f(x)^2 - f(y)^2| > 2M(M_i - m_i).$$

But for any $x, y \in [x_{i-1}, x_i]$ we have

$$|f(x)^2 - f(y)^2| \leq 2M |f(x) - f(y)| \leq 2M(M_i - m_i),$$

so the desired inequality holds.

Finally, assume that f is integrable and fix $\epsilon > 0$. Then by the Darboux criterion, there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \frac{\epsilon}{2M}$. It follows that

$$U(f^2, P) - L(f^2, P) = \sum_{i=1}^n (M'_i - m'_i) \Delta x_i \leq 2M \sum_{i=1}^n (M_i - m_i) \Delta x_i < 2M \cdot \frac{\epsilon}{2M} = \epsilon,$$

whence f^2 is integrable by the Darboux criterion.

(b) Note that since $(f + g)^2 \geq 0$, $(f - g)^2 \geq 0$, we have

$$\int (f + g)^2 \geq 0 \quad \text{and} \quad \int (f - g)^2 \geq 0.$$

Expanding these integrals using the basic properties of integration and rearranging, we obtain

$$-2 \int fg \leq \int f^2 + \int g^2 \leq 2 \int fg,$$

or equivalently $2|\int fg| \leq \int f^2 + \int g^2$ as needed.

(c) We begin by considering the case where $\int f^2 = 0$ or $\int g^2 = 0$. We claim that this implies that $\int fg = 0$, which will yield the Cauchy-Schwarz inequality in this case, since then both sides vanish. The situation is symmetrical in f and g , so it suffices to treat the case $\int g^2 = 0$. If $\int f^2 = 0$ then the claim follows immediately, so we can also assume that $\int f^2 \neq 0$. Fix $\epsilon > 0$ and consider the function $g_0 := \frac{1}{2\epsilon}(\int f^2)g$. Note that g_0 is a scalar multiple of g , so we have

$$\int g_0^2 = \left(\frac{1}{2\epsilon} \int f^2\right)^2 \int g^2 = 0.$$

Applying part (b) to f and g_0 , we obtain that

$$\frac{1}{\epsilon} \left(\int f^2\right) \left|\int fg\right| = \left|\int fg_0\right| \leq \int f^2 + \int g_0^2 = \int f^2$$

and hence $|\int fg| \leq \epsilon$. Since ϵ was arbitrary, it follows that $\int fg = 0$ as claimed.

Finally, we consider the case where $\int f^2, \int g^2 \neq 0$. Put $f_0 := (\int f^2)^{-\frac{1}{2}}f$ and $g_0 := (\int g^2)^{-\frac{1}{2}}g$ as in the hint. Applying part (b) to f_0 and g_0 , we obtain

$$\begin{aligned} 2 \left(\int f^2\right)^{-\frac{1}{2}} \left(\int g^2\right)^{-\frac{1}{2}} \left|\int fg\right| &= 2 \left|\int f_0 g_0\right| \leq \int f_0^2 + \int g_0^2 \\ &= \left(\int f^2\right)^{-1} \left(\int f^2\right) + \left(\int g^2\right)^{-1} \left(\int g^2\right) = 2. \end{aligned}$$

Rearranging yields

$$\left|\int fg\right| \leq \left(\int f^2\right)^{\frac{1}{2}} \left(\int g^2\right)^{\frac{1}{2}},$$

and after squaring we obtain the Cauchy-Schwarz inequality.

5. First, we claim that for any partition P of $[0, 1]$, we have $L(f, P) = 0$. Namely, since the irrational numbers are dense in $\mathbb{R}$, for any $1 \leq i \leq n$ there exists $x \in [x_{i-1}, x_i]$ such that $x \notin \mathbb{Q}$. Then $f(x) = 0$, so we see that $m_i = 0$ for all $1 \leq i \leq n$ and hence $L(f, P) = 0$ as claimed.

To show that $\int_0^1 f(x) dx = 0$, it therefore suffices to show that for any $\epsilon > 0$, there exists a partition P such that $U(f, P) < \epsilon$. For any $n \in \mathbb{N}$, let P_n denote the partition of $[0, 1]$ into n equal parts. We claim that for any $M > 0$, there exist only finitely many $x \in [0, 1]$ such that $f(x) \geq M$ (here “finitely many” allows for the possibility that there exist no such x). If $x \notin \mathbb{Q}$, then $f(x) = 0 < M$, so we need only consider rational numbers. Certainly there are only finitely many $q \in \mathbb{N}$ such that $\frac{1}{q} \geq M$, or

equivalently $q \leq \frac{1}{M}$. For any such q , if $p \in \mathbb{N}$ satisfies $\frac{p}{q} \in [0, 1]$, then we must have $p \leq q$, and there are only finitely many such numbers p , whence the claim.

Now apply the claim with $M = \frac{\epsilon}{2}$, so we conclude that there are only finitely many $x \in [0, 1]$ such that $f(x) \geq \frac{\epsilon}{2}$. Denote the finite set of such numbers by $S \subset [0, 1]$, and let $m := \#S$. Choose $n \in \mathbb{N}$ so that $n > \frac{4m}{\epsilon}$, and let

$$I := \{1 \leq i \leq n \mid \text{there exists } x \in S \cap [x_{i-1}, x_i]\}.$$

Note that since any $x \in S$ can belong to $[x_{i-1}, x_i]$ for at most two values of i , we have $\#I \leq 2m$. Finally, we observe that for any $1 \leq i \leq n$ we have $M_i \leq 1$, and for $i \notin I$ we have $M_i \leq \frac{\epsilon}{2}$. Now we compute

$$\begin{aligned} U(f, P) &= \sum_{i=1}^n M_i \Delta x \\ &= \frac{1}{n} \sum_{i=1}^n M_i \\ &= \frac{1}{n} \left(\sum_{i \in I} M_i + \sum_{i \notin I} M_i \right) \\ &\leq \frac{1}{n} \left(\sum_{i \in I} 1 + \sum_{i \notin I} \frac{\epsilon}{2} \right) \\ &\leq \frac{1}{n} \left(2m + n \cdot \frac{\epsilon}{2} \right) \\ &= \frac{2m}{n} + \frac{\epsilon}{2} \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$